S Naga Surya

AI / Backend Engineer | Agentic Systems | FastAPI | RAG

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Professional Summary

AI / Backend Engineer with 2+ years of experience building agentic and multi-agent workflows, RAG systems, and asynchronous FastAPI services using LangGraph, LangChain, MCP, Anthropic Claude, OpenAI, and Playwright. Delivered secure enterprise systems across autonomous testing, product planning, software-delivery orchestration, and logistics document intelligence.

Technical Skills

Backend Engineering: Python, SQL, FastAPI, asynchronous services, REST APIs, Socket.IO (WebSocket streaming)

Agentic AI & LLMs: LangGraph, LangChain, MCP (FastMCP), Anthropic Claude API, OpenAI API, multi-agent orchestration, tool calling, multi-turn agent loops, prompt engineering, prompt caching, guardrails

Retrieval & Data: RAG, Pinecone, FAISS, MongoDB, PostgreSQL, Supabase

Automation & Applied AI: Playwright, Azure Document Intelligence, Tavily, pytest

Cloud & Delivery: Docker, Git, CI/CD (GitHub Actions), Google Cloud Run Jobs, AWS (RDS, S3, CloudWatch)

Professional Experience

Software Engineer **Jul 2026 – Present**

Ideyalabs Tech LLP, Hyderabad

Apex MyDaylight — Logistics Client Portal & Document Intelligence

Tech Stack: Python, FastAPI, PostgreSQL, Azure Document Intelligence, AWS

- Designed an Azure Document Intelligence pipeline to classify invoices, shipping labels, Bills of Lading, and related logistics documents.
- Routed each document type to custom extraction, fixed-template, or prebuilt read models to produce structured operational data.
- Build asynchronous FastAPI services with Pydantic validation for shipment visibility, rate quotes, accounts, and customer-service workflows.
- Develop React and python modules and secure integrations with internal logistics systems, supported by role-based access control and audit logging with features including shipments and rate quotes.
- Support modernization from legacy TruckMate/.NET systems to Python services by making historical document data available to new workflows.
- Optimize PostgreSQL queries, indexes, and connection pooling while supporting AWS-hosted components across RDS, S3, and CloudWatch.

Associate Software Engineer **Jan 2025 – Jun 2026**

Ideyalabs Tech LLP, Hyderabad

PO Agent — Agentic Product Planning Platform

Tech Stack: Python, LangGraph, MongoDB, Tavily, RAG

- Built the foundational LangGraph orchestration and stateful workflow for a multi-stage, agentic product-planning platform.
- Converted an initial product idea into structured research, product scope, technical documentation, and release-planning stages.
- Owned a Tavily-enabled research workflow that gathers current market context and supporting evidence from live web sources.
- Generated competitor analysis, expected product workflows, and SWOT analysis to establish the opportunity and product direction.
- Transformed research into MVP feature hierarchies, user stories, sub-features, expected behavior, and delivery requirements.
- Orchestrated PRDs, architecture flowcharts, feature mind maps, and phased release plans; persisted state in MongoDB and supported RAG retrieval.

Eco Online — Agent-Driven Software Delivery Platform
Tech Stack: Python, FastAPI, LangGraph, MCP, Supabase, GitHub, Anthropic Claude, OpenAI

- Owned the FastAPI and LangGraph backend for a client platform combining PO Agent and ITAF into an agent-monitored delivery lifecycle.
- Built MCP integrations to fetch Jira tickets, acceptance criteria, repository context, pull-request state, and deployment information.
- Created workflows that refine ticket scope and acceptance criteria before passing implementation tasks to AI coding agents.
- Orchestrated GitHub pull-request creation and monitored merge and deployment state without requiring manual workflow coordination.
- Triggered ITAF after deployment to generate and execute live tests mapped directly to each ticket's acceptance criteria.
- Integrated Anthropic Claude and OpenAI using tool calling, multi-turn loops, and prompt caching; stored workflow data in Supabase.

Software Engineering Intern **Jun 2024 – Dec 2024**
Ideyalabs Tech LLP, Hyderabad

ITAF — Ideyalabs Testing Automation Framework
Tech Stack: Python, LangGraph, LangChain, Playwright, FastMCP, Socket.IO, Pinecone, PostgreSQL, MongoDB

- Built an AI testing platform for two enterprise customers, reducing required manual and automation testing effort by approximately 60% versus conventional regression cycles and shifting testers toward validation and audit.
- Developed LangChain and LangGraph workflows that convert Jira user stories and ticket descriptions into executable plain-English test steps.
- Used Playwright headless browsers, rendered-screen context, and normalized DOM snapshots to discover features and understand the current UI state.
- Built reasoning-and-action execution loops with per-test data generation to run test cases sequentially against live applications.
- Implemented LLM-powered self-healing that updates only the affected interactive element while preserving the validated test flow.
- Converted successful runs into reusable Playwright scripts or English steps and expanded them into positive and negative scenarios.
- Added failure analysis for ticket drift, application defects, and unrecoverable errors, returning evidence-backed root-cause information.
- Built a guarded RAG assistant for ITAF guidance and testing access, using FastMCP and Socket.IO for tools and real-time updates.

Education

KL University, Andhra Pradesh *2021 – 2025*
B.Tech in Artificial Intelligence & Data Science CGPA: 9.62

Certifications

Google TensorFlow Developer Certificate | IBM Python | Salesforce AI Associate (ID: 5591179)